

5.3.25

AI25BTECH110031
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Question(5.3.25) Solve the following pair of linear equations

$$3x + 4y = 10 \quad (0.1)$$

$$2x - 2y = 2 \quad (0.2)$$

Solution:

Given equations:

$$\begin{pmatrix} 3 & 4 \end{pmatrix} \mathbf{x} = 10 \quad (0.3)$$

$$\begin{pmatrix} 2 & -2 \end{pmatrix} \mathbf{x} = 2 \quad (0.4)$$

In augmented matrix form:

$$\left(\begin{array}{cc|c} 3 & 4 & 10 \\ 2 & -2 & 2 \end{array} \right) \quad (0.5)$$

$$R_2 \rightarrow R_2 - \frac{2}{3}R_1 \Rightarrow \left(\begin{array}{cc|c} 3 & 4 & 10 \\ 0 & -\frac{14}{3} & -\frac{14}{3} \end{array} \right) \quad (0.6)$$

$$R_2 \rightarrow -\frac{3}{14}R_2 \Rightarrow \left(\begin{array}{cc|c} 3 & 4 & 10 \\ 0 & 1 & 1 \end{array} \right) \quad (0.7)$$

$$R_1 \rightarrow R_1 - 4R_2 \Rightarrow \left(\begin{array}{cc|c} 3 & 0 & 6 \\ 0 & 1 & 1 \end{array} \right) \quad (0.8)$$

$$R_1 \rightarrow \frac{1}{3}R_1 \Rightarrow \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right) \quad (0.9)$$

Hence,

$$x = 2, \quad y = 1. \quad (0.10)$$

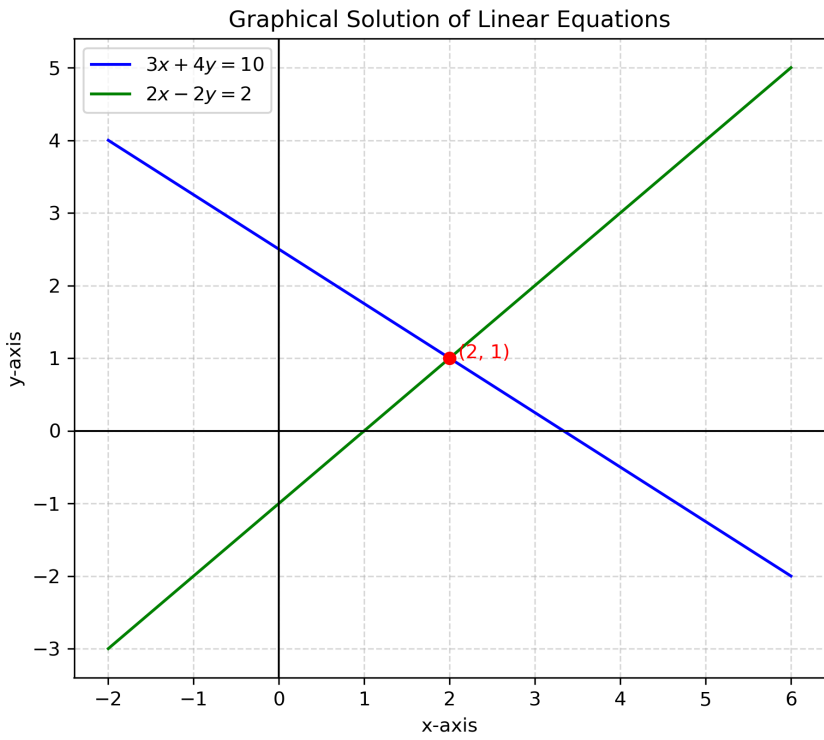


Fig. 0.1